

SHETH L.U.J AND SIR M.V. COLLEGE
SUBJECT NAME: Artificial Intelligence

Practical 3

Aim: Decision Tree Learning

- Implement the Decision Tree Learning algorithm to build a decision tree for a given dataset.
- Evaluate the accuracy and effectiveness of the decision tree on test data.
- Visualize and interpret the generated decision tree.

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CODE:

```
T074 Ai Pract 3 Tree.py - C:/Users/PRIYANKA/Desktop/T074 Ai/T074 Ai Pract 3 Tree.py (3.13.1)
File Edit Format Run Options Window Help

import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier, plot_tree
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix

df = pd.read_csv('GamingStudy_data.csv', encoding='latin-1')

features = ['Hours', 'Age', 'SWL_T', 'SPIN_T', 'Narcissism']
target = 'GAD_T'

df_clean = df.dropna(subset=features + [target]).copy()

X = df_clean[features]
y = (df_clean[target] > 9).astype(int)

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42)

clf = DecisionTreeClassifier(criterion='gini', max_depth=3, random_state=42)
clf.fit(X_train, y_train)

y_pred = clf.predict(X_test)

print("--- Model Evaluation ---")
print("Accuracy: {accuracy_score(y_test, y_pred):.4f}")
print("\nClassification Report:")
print(classification_report(y_test, y_pred, target_names=['Low Anxiety (<=9)', 'High Anxiety (>9)']))

print("Confusion Matrix:")
print(confusion_matrix(y_test, y_pred))

plt.figure(figsize=(16, 10))
plot_tree(
    clf,
    feature_names=features,
    class_names=['Low Anxiety', 'High Anxiety'],
    filled=True,
    rounded=True,
    fontsize=11,
    proportion=True
)
plt.title("Decision Tree: Predicting High Anxiety in Gamers", fontsize=16)
plt.tight_layout()
plt.show()
```

OUTPUT:

```
*IDLE Shell 3.13.1*
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Python 3.13.1 (tags/v3.13.1:0671451, Dec 3 2024, 19:06:28) [MSC v.1942 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.

>>>
===== RESTART: C:/Users/PRIYANKA/Downloads/123.py =====
--- Model Evaluation ---
Accuracy: 0.8315

Classification Report:
precision    recall  f1-score   support

Low Anxiety (<=9)    0.87    0.94    0.90    3199
High Anxiety (>9)    0.48    0.28    0.36    634

 accuracy          0.83    3833
 macro avg         0.68    0.61    0.63    3833
 weighted avg      0.81    0.83    0.81    3833

Confusion Matrix:
[[3007 192]
 [454 180]]
```

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